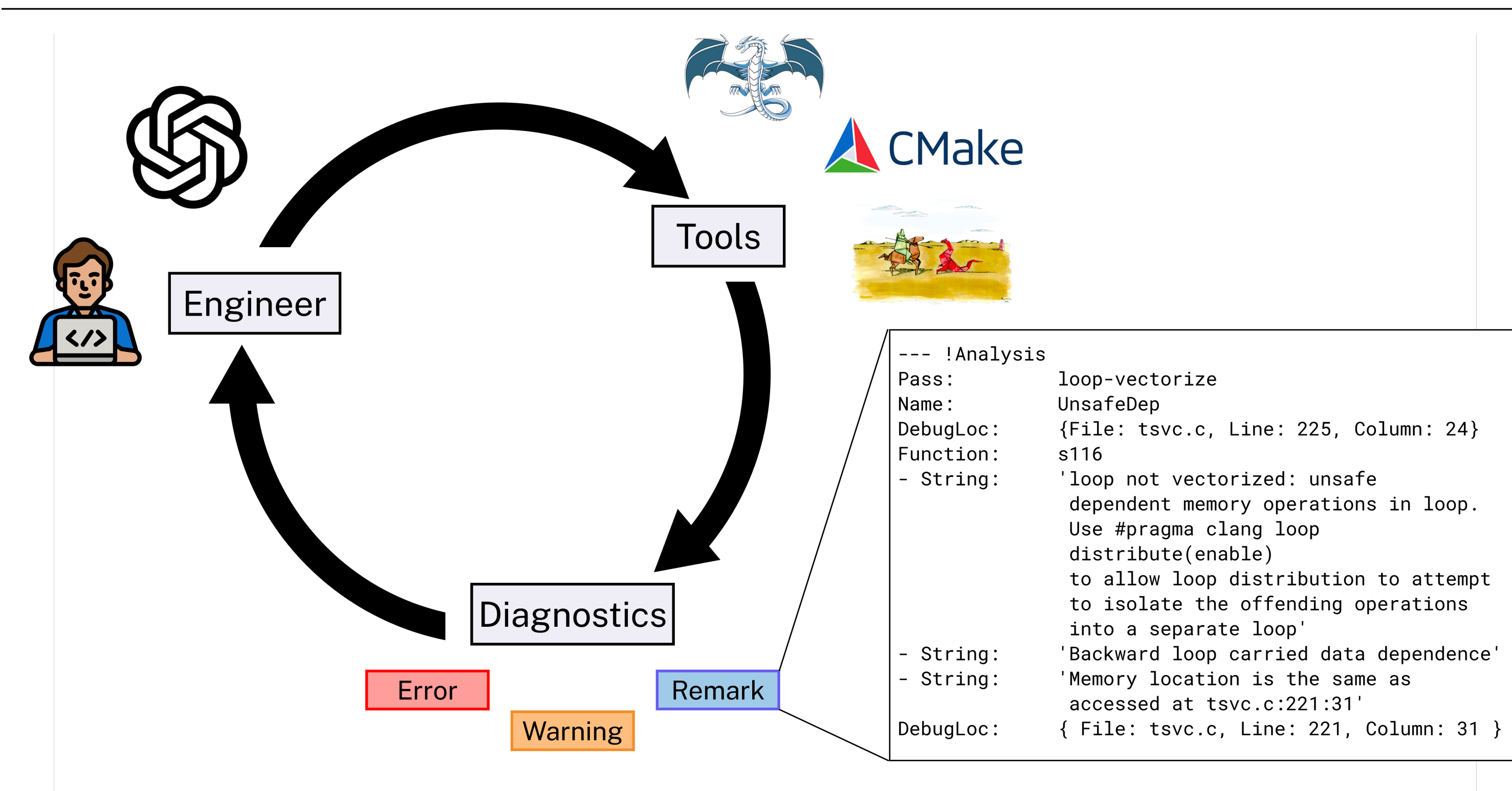


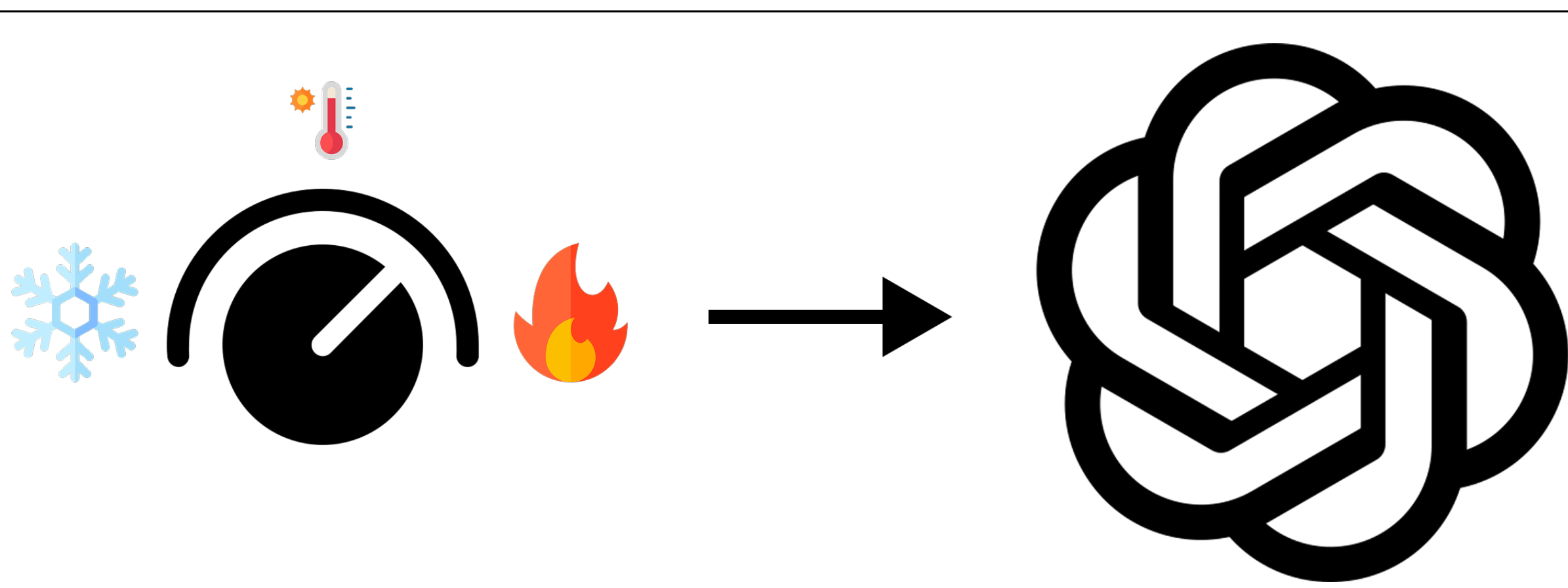
How To Evaluate Tool Diagnostics Using AI

Akash Deo, Simone Campanoni, Tommy McMichen

Engineers use tool diagnostics everyday.



It's hard to improve tool diagnostics, but AI can help.

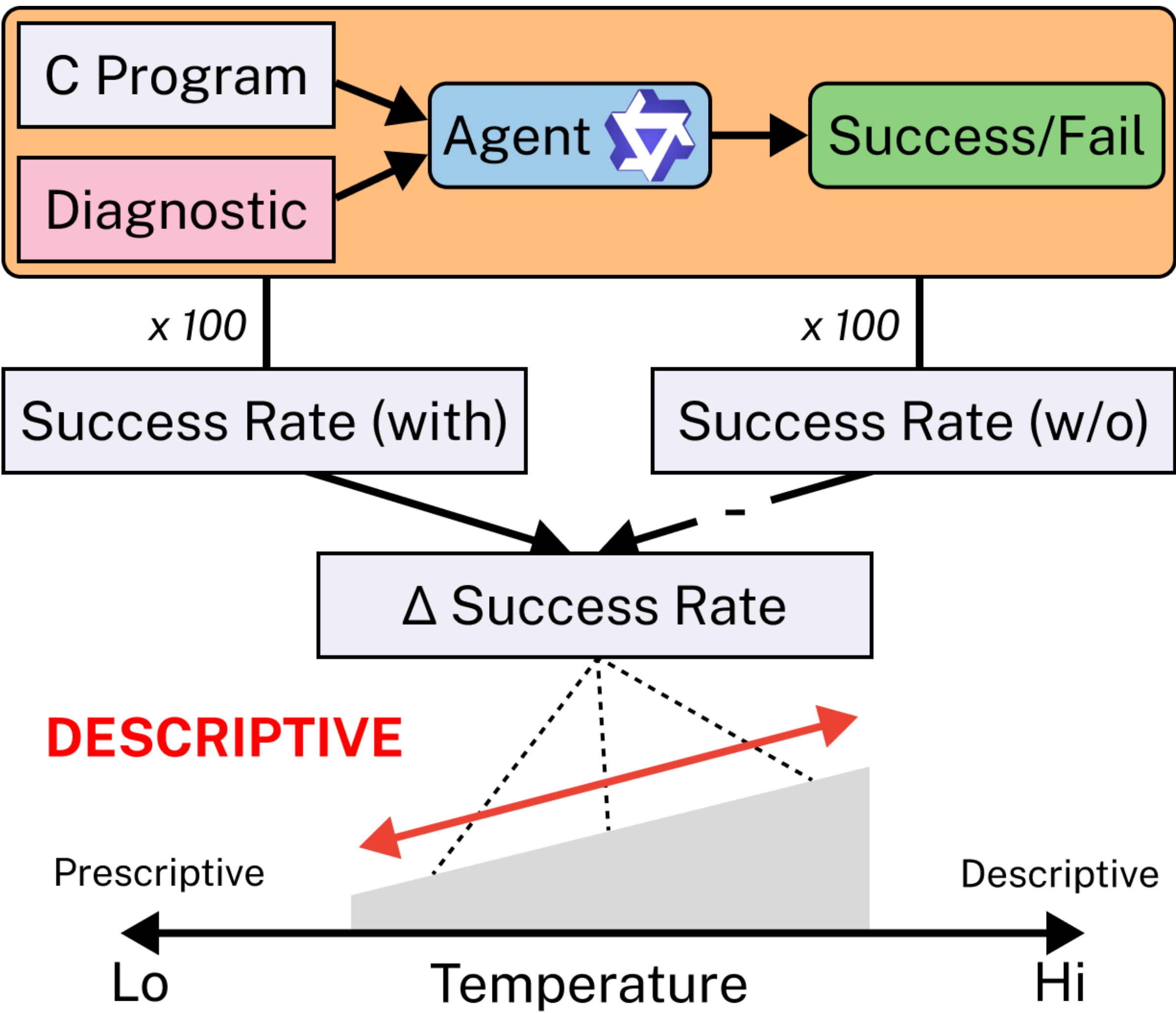


Descriptive: What is the problem?

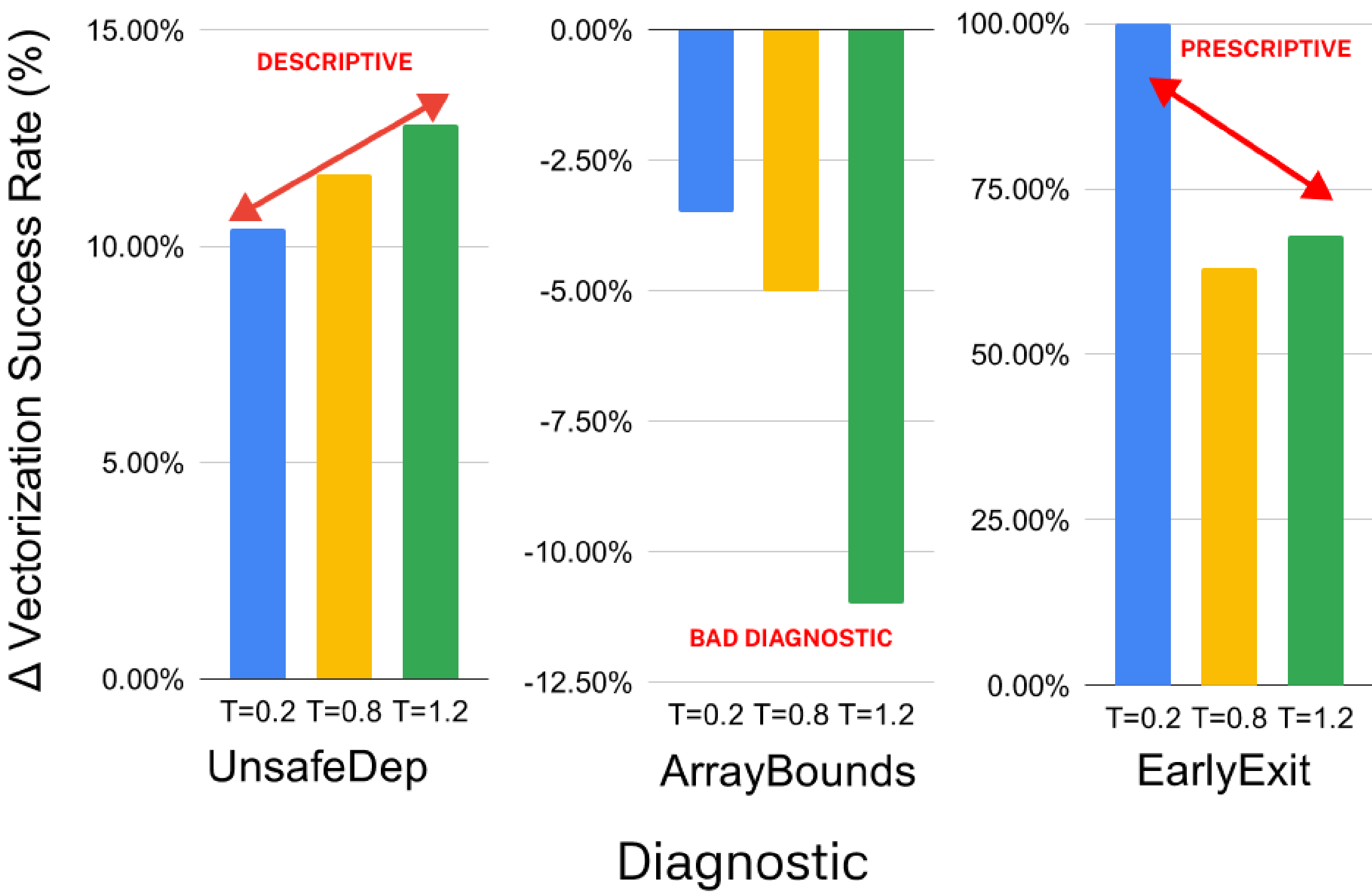
Prescriptive: How do you solve it?

Can we automatically measure how descriptive and prescriptive our tool diagnostics are?

CANARY measures how different diagnostics perform over different model temperatures.



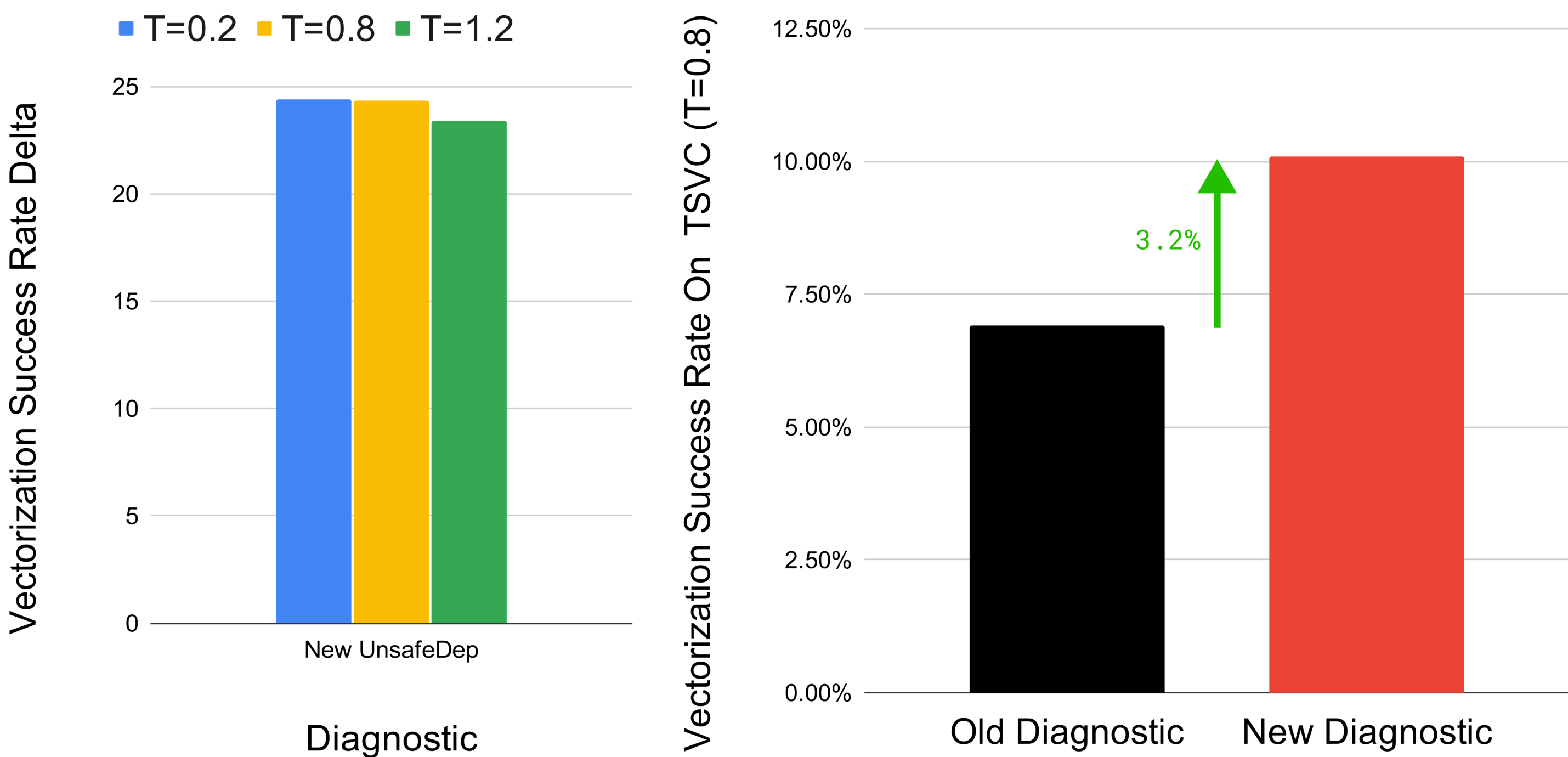
Sweeping over temperature tells us whether a diagnostic is descriptive or prescriptive.



Using CANARY helps improve tool diagnostics.

Case Study: LLVM Vectorizer Diagnostics

```
tsvc.c:220:9: Dependence report for loop in s116
Status: Not vectorizable
Backward write-after-read dependence between source (tsvc.c:221:18) and destination (tsvc.c:225:24)
Forward write-after-read dependence between source (tsvc.c:224:24) and destination (tsvc.c:225:22)
Forward write-after-read dependence between source (tsvc.c:223:24) and destination (tsvc.c:224:22)
Forward write-after-read dependence between source (tsvc.c:222:24) and destination (tsvc.c:223:22)
Forward write-after-read dependence between source (tsvc.c:221:20) and destination (tsvc.c:222:22)
```



Do You Have Feedback?

1. Do you know any other use cases for CANARY?
2. What other info would you want to know about tool diagnostics?

CoDAIM '26 Paper

